

Practice Programs for OOPS

Interfaces

1. Create an interface Shape with a method calculateArea(). Implement it in a Rectangle class.
2. Create an interface Vehicle with methods start() and stop(). Implement the interface in a Car class.
3. Create an interface Animal with method sound(). Implement it in Dog and Cat classes.
4. Create an interface Bank with method getInterestRate(). Implement it in two bank classes.
5. Create an interface Printable with method print(). Implement it in a class that displays student details.
6. Create an interface Remote with methods on() and off(). Implement it in a TV class.
7. Create an interface Calculator with method add(). Implement it in a class that adds two numbers.
8. Create an interface Sports with method play(). Implement it in a Cricket class.
9. Create an interface Payment with method payBill(). Implement it in a CreditCard class.
10. Create an interface Employee with method displayRole(). Implement it in Manager and Developer classes.

Exception Handling

1. Write a program to handle division by zero using try-catch.
2. Create a program that handles ArrayIndexOutOfBoundsException.
3. Write a program to handle invalid number input using NumberFormatException.
4. Create a program using multiple catch blocks.
5. Write a program demonstrating the use of finally block.
6. Create a custom exception called InvalidAgeException.
7. Write a program to check voting eligibility using exception handling.
8. Create a program that throws an exception when password length is less than 6.

9. Write a program to handle NullPointerException.
10. Create a program that accepts marks and throws an exception if marks are negative.

Collection Framework

1. Create an ArrayList to store student names and display them.
2. Write a program to add and remove elements from an ArrayList.
3. Create a LinkedList of cities and display all cities.
4. Write a program using Vector to store integer values.
5. Write a program to search an element in an ArrayList.
6. Create a program to count total elements in a collection.
7. Write a program to iterate through an ArrayList using loop.

File Handling

1. Write a program to create a text file.
2. Create a program to write student details into a file.
3. Write a program to read data from a text file.
4. Create a program to count number of lines in a file.
5. Write a program to append text into an existing file.
6. Create a program to copy contents from one file to another.
7. Write a program to display file information such as name and size.
8. Create a program to search a word in a file.
9. Write a program to store employee details in a file.
10. Create a program to read and display all words from a file.

Multithreading

1. Create a thread using Thread class.
2. Create a thread using Runnable interface.
3. Write a program that displays numbers from 1 to 10 using a thread.
4. Create two threads that display different messages.
5. Write a program demonstrating thread priority.
6. Create a thread that prints current time every second.
7. Write a program using sleep() method in threads.
8. Create a program to check whether a thread is alive.
9. Write a program demonstrating multithreading with two tasks.
10. Create a program that starts and stops a thread.

GUI Programming using Swing

1. Create a GUI with a button that displays a message.
2. Design a login form using Swing.
3. Create a calculator to add two numbers.
4. Design a GUI to change background color using buttons.
5. Create a form to display student details using JOptionPane.
6. Design a GUI with radio buttons for gender selection.
7. Create a GUI with checkboxes for hobbies.
8. Design a program to convert text to uppercase.
9. Create a GUI with combo box for country selection.
10. Design a GUI with a reset button to clear all fields.

GUI Application using Database Connectivity

1. Create a Swing form to insert student details into database.
2. Design a login form that validates user credentials from database.
3. Create a GUI to display all employee records from database.
4. Design a GUI to search student details using Roll Number.
5. Create a GUI to update employee salary in database.
6. Design a Swing application to delete student record from database.
7. Create a GUI to insert book details into database.